

HWO ExoEarth Yields for a Mass-Selected Planet Sample in a Narrowed Habitable Zone

A rederivation of the AYO yield model of Stark et al., laid out as a probabilistic model

Prepared in the `yields` vaibify project

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Abstract

We ask how many exoEarth candidates the baseline Habitable Worlds Observatory (HWO) mission of Stark et al. (2024) would find if the target population were planets of $0.5\text{--}2\,M_{\oplus}$ (with $R = M^{0.27}$) in a habitable zone of 0.96–1.20 AU scaled by $\sqrt{L_{\star}}$, rather than the canonical $R \leq 1.4\,R_{\oplus}$ planets in 0.95–1.67 AU. Answering it requires re-running the survey optimization, because the survey reallocates its time as the expected number of detections changes. We rederive the Altruistic Yield Optimization (AYO) model from the published equations and parameters, and write it down as a probabilistic graphical model whose single decision node is the survey allocation. With the extended-source throughput T_{sky} treated as a function of separation, as Stark et al. (2019) define it, the model reproduces Stark’s published 22.5 candidates to 0.5% with no fitted parameter; the one throughput factor it carries, previously 1.62, is now $\kappa = 0.988$. Exozodi is a random variable of the pipeline, marginalized over 200 draws with the survey re-optimized for each. The model gives a median canonical yield of 19.3 (90% interval 5.9–51.6) and a median redefined yield of 4.6 (1.3–15.5). The yield ratio is $0.245^{+0.035}_{-0.025}$, larger than the occurrence ratio of 0.204 because the redefined box removes the faintest planets. The probability of reaching 25 candidates falls from 35.7% to 1.19%. Of 23 independent checks against the published record, 19 agree, and nine of Stark’s figures are reproduced in their own axes beside the originals. The main remaining disagreement is that this model’s spectral characterization times depend on target difficulty and aperture more steeply than Stark’s. A per-star table gives every target’s Earth zone and classic habitable zone and what the survey expects to find in each.

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1 The question and the answer

The request was to trace the ExoVista/AYO yield chain back to its assumptions about the planet population, couple it to a Bayesian sampler, and predict how many planets of $0.5\text{--}2 M_{\oplus}$, $R = M^{0.27}$, around FGK stars, the baseline HWO mission would find if the habitable zone were $0.96\text{--}1.20$ AU for the Sun, scaled as $\sqrt{L_{\star}}$ for other stars. The answer is summarized in Table 1.

	Canonical box	Redefined box
Habitable zone (AU, scaled by $\sqrt{L_{\star}}$)	0.95–1.67	0.96–1.20
Planet selection	$0.8 (a/\text{EEID})^{-1/2} < R/R_{\oplus} \leq 1.4$	$0.5 \leq M/M_{\oplus} \leq 2, R = M^{0.27}$
Occurrence η (median, 68%)	$0.255^{+0.282}_{-0.134}$	$0.0520^{+0.0577}_{-0.0274}$
Realized yield, median	19.3	4.6
Realized yield, mean	22.5	6.0
Realized yield, 90% interval	5.9–51.6	1.3–15.5
$P(\geq 25 \text{ candidates})$	35.7%	1.19%
Occurrence ratio, redefined/canonical	$0.204^{+0.017}_{-0.016}$	
Yield ratio, redefined/canonical	$0.245^{+0.035}_{-0.025}$	

Table 1: The answer for the baseline 6 m (inscribed) LUVOIR-B-like design. Yields are realized counts, marginalized over the $\eta_{\oplus}$ uncertainty, the planet draw, the geometric albedo and the survey’s re-optimization; they are the quantity Stark et al. (2024) plot in Fig. 10 and compute P_{25} from.

The yield ratio is the robust result. It is taken at one aperture, inside one implementation, with the same calibration in numerator and denominator. Across the model revisions of the final sessions

(exozodi, $\eta_{\oplus}$ width, the separation-dependent T_{sky} , and exozodi marginalized over draws) it has moved only from 0.246 to 0.241 to 0.245, inside its own error bar. The absolute yields are conditional on the open problem of Sec. 8.

Why the yield ratio exceeds the occurrence ratio (Sec. 6.1) is itself a result: the redefined box keeps the brighter, closer-in planets, frees characterization time, and loses some distant stars, and the first two effects win. A rescaling of Stark’s yield by the occurrence ratio alone would have been wrong by 20%, in a direction that cannot be predicted without running the optimization.

2 The model as a probabilistic graphical model

2.1 Why a graphical model

The AYO yield calculation is usually described procedurally: compute completeness, optimize exposure times, count planets, then add uncertainty sources one at a time. That description hides the structure that decides every number in this report — which quantities are *random*, which are *fixed*, which are *fitted*, and, crucially, which random quantities the survey planner is allowed to see when it allocates its time. Stark et al. (2024, Sec. 3) call the last distinction *actionable* versus *static* uncertainty. In a graphical model it has a precise meaning, and it is the single most consequential modelling choice after the radiometry.

Figure 1 shows the model as an influence diagram: a directed acyclic graph of random variables with one *decision* node, the survey allocation. Every node is implemented by a named function in the pipeline, listed in Table 2.

2.2 The nodes

Node	Type	Specification	Source; step
$\eta_{\oplus}$	random, global	lognormal, median 0.257, ln-width 0.76 (Sec. 3.2)	Stark+2024 Sec. 3.5, read at $z = 1$; A06
α, β	random, global	SAG13 shape; posterior of an <code>emcee</code> chain, used only through $\rho_{\mathcal{B}}$	SAG13; A06
$\rho_{\mathcal{B}}$	deterministic	$\eta_{\mathcal{B}}/\eta_{\oplus}$, the SAG13 integral over $\mathcal{B}$ relative to the canonical box; independent of Γ	A06
$\eta_{\mathcal{B}}$	deterministic	$\eta_{\oplus} \rho_{\mathcal{B}}$	A06
$\mathbf{s}_i$	observed	$d, L_{\star}, T_{\text{eff}}, R_{\star}$, HIP id	HPIC (Tuchow+2024); A01
z_i	random, per star	exozodi level, $z_i \sim p_z$; four stars fixed at LBTI values; 200 joint draws of $\mathbf{z}$	Stark+2024 Sec. 3.3, Fig. 9; A02, A04
C_i	deterministic	planner’s completeness $C_i(\tau, k)$ at $A_G = 0.2$, over exposure and visits, counting only planets that are also characterizable	Stark+2019; A04
λ	deterministic	equal-slope multiplier closing the two-year budget	Stark+2014; A05
$\mathbf{a}_i$	decision	(τ_i, k_i) : the vertex of star i ’s concave cost envelope at slope λ	A05, A07
N_i	random, per star	$N_i \sim \text{Poisson}(\eta_{\mathcal{B}})$	Stark+2024 Sec. 3.1; A07
$\mathbf{o}_{ij}$	random, per planet	circular orbit: a, R from SAG13 over $\mathcal{B}$; $\cos i \sim \mathcal{U}$; independent epoch per visit	ExoVista/AYO; A04
A_{ij}	random, per planet	geometric albedo, $\mathcal{U}(0.08, 0.32)$	Stark+2024 Sec. 3.2; A04
τ_{ij}	deterministic	detection time per visit (two channels in quadrature, S/N 7) and characterization time (R 140, best of nine bandpasses, best orbital phase)	Stark+2019 Eqs. 1–9; A04
D_{ij}	deterministic	$\mathbb{K}[\min_{v \leq k_i} \tau_{ijv}^{\text{det}} \leq \tau_i \wedge \tau_{ij}^{\text{char}} \leq 2 \text{ mo}]$	Stark+2019 Sec. 3; A04
Y	deterministic	$\sum_{i,j} D_{ij}$	A07
κ	fitted	throughput factor on every band, fitted to Y^{pub} : 0.988	A03

Table 2: Every node of Fig. 1, what it is, and where it comes from.

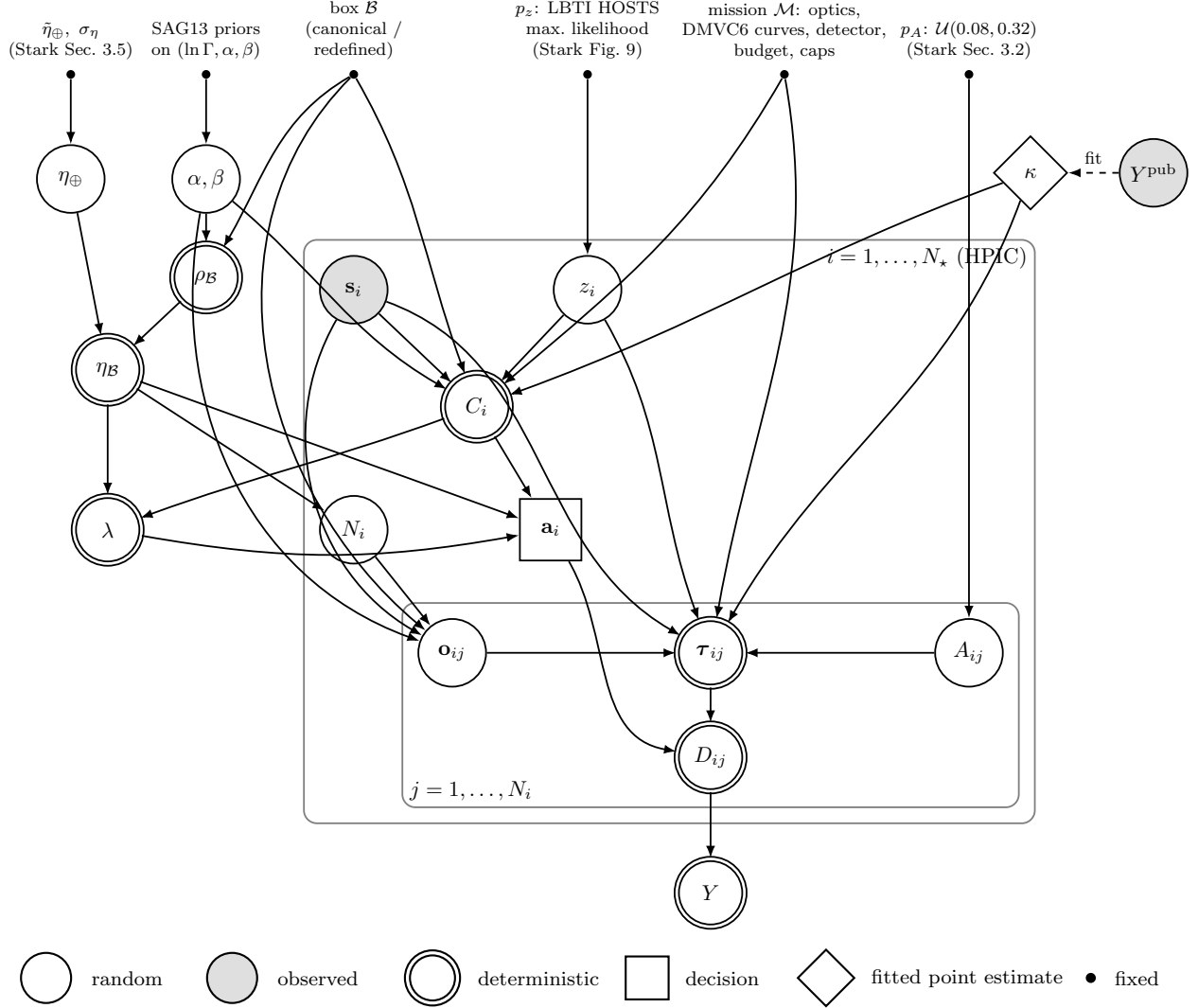


Figure 1: The yield model as an influence diagram. Outer plate: the $N_\star$ screened HPIC stars; inner plate: the N_i planets actually present around star i . The survey allocation $\mathbf{a}_i = (\tau_i, k_i)$ — exposure per visit and number of visits — is the one decision node; it is set by the equal-slope multiplier λ from the planner’s completeness curves C_i and the box occurrence rate $\eta_{\mathcal{B}}$. A planet is counted ($D_{ij} = 1$) when its detection time at some visit is within the allocation and its spectral characterization time is within the two-month cap; $Y = \sum_{ij} D_{ij}$. The throughput factor κ is not a random variable: it is fitted once, at fixed exozodi, to the published expected yield $Y^{\text{pub}} = 22.5$ (dashed), and comes out at 0.988, so the edge carries almost no information. Four stars with LBTI-measured exozodi have z_i observed rather than drawn (not shown). The edges that decide the uncertainty budget are the ones *into* $\mathbf{a}_i$: $\eta_\oplus$ and z_i reach it (actionable), A_{ij} and N_i do not (static).

2.3 The joint distribution, and how the pipeline marginalizes it

Conditioning on the fixed quantities $(\mathcal{M}, \mathcal{B}, \kappa, \{\mathbf{s}_i\})$, the graph factorizes as

$$p(\eta_{\oplus}, \alpha, \beta, \mathbf{z}, \mathbf{N}, \mathbf{o}, \mathbf{A}, Y) = p(\eta_{\oplus}) p(\alpha, \beta) \prod_i p_z(z_i) \text{Poisson}(N_i; \eta_{\mathcal{B}}) \\ \times \prod_{j=1}^{N_i} p(\mathbf{o}_{ij} \mid \alpha, \beta, \mathcal{B}, \mathbf{s}_i) p_A(A_{ij}) \cdot \delta\left(Y - \sum_{ij} D_{ij}\right), \quad (1)$$

with the allocation entering through D_{ij} as the deterministic function

$$\mathbf{a}_i = \mathbf{a}_i(\lambda(\eta_{\mathcal{B}}, \mathbf{C}), C_i), \quad C_i = C_i(\mathbf{s}_i, z_i, \mathcal{M}, \kappa, \mathcal{B}, \alpha, \beta). \quad (2)$$

Two properties make this tractable, and the pipeline uses both.

Poisson thinning. Given the allocation, planets are counted independently with a probability that depends only on the star. Marginalizing the planet draw inside the plate is then exact: if $N_i \sim \text{Poisson}(\eta_{\mathcal{B}})$ and each planet is counted with probability $\bar{C}_i$ — the completeness at the allocated vertex, averaged over the albedo draw — the counted planets are $\text{Poisson}(\eta_{\mathcal{B}} \bar{C}_i)$, and the survey total is

$$Y \mid \eta_{\oplus}, \mathbf{z} \sim \text{Poisson}\left(\Lambda(\eta_{\oplus}, \mathbf{z})\right), \quad \Lambda = \eta_{\mathcal{B}} \sum_i \bar{C}_i(\mathbf{a}_i(\eta_{\mathcal{B}}, \mathbf{z})). \quad (3)$$

Step A07 evaluates Λ on a 24-node grid in $\eta_{\oplus}$, *re-optimizing the survey at every node*, interpolates it, and draws one Poisson count per posterior sample of $\eta_{\oplus}$ (14,400 samples). This is Stark’s construction exactly: his Sec. 3.1 draws a Poisson number of planets per star and counts those with a valid visit record.

Albedo is static, so it enters $\bar{C}_i$ and not $\mathbf{a}_i$. The planner allocates against C_i computed at $A_G = 0.2$; the count is then read from the same planets and orbits with A_{ij} drawn. A darker-than-planned planet can fail the fixed allocation; a brighter one gains little. This asymmetry is the whole of the albedo penalty, and the graph makes it unavoidable: A_{ij} has no path into $\mathbf{a}_i$.

What is marginalized, and what is not yet.

- $\eta_{\oplus}$: marginalized, with the survey re-optimized per value (actionable).
- $N_i, \mathbf{o}_{ij}, A_{ij}$: marginalized exactly by thinning, Eq. (3).
- α, β : marginalized for the *ratio* $\rho_{\mathcal{B}}$; the planet population inside each box uses the SAG13 central values.
- $\mathbf{z}$: marginalized over 200 joint draws (Stark uses 500), with the survey *re-optimized for every draw*, because exozodi is actionable. A04 tabulates each star’s completeness on a 23-level exozodi grid (0–1024 zodis) and each draw interpolates between levels; against direct evaluation at the drawn levels this changes the optimized yield by at most 0.24%. Each posterior sample of $\eta_{\oplus}$ is paired with one draw of $\mathbf{z}$. Over the draws the fixed- $\eta_{\oplus}$ yield is 18.30 with a draw-to-draw scatter of 0.57.

2.4 Actionable and static, stated precisely

A source of uncertainty is *actionable* if its node has a directed path into the decision node, and *static* if it does not. In Fig. 1:

- $\eta_{\oplus} \rightarrow \eta_{\mathcal{B}} \rightarrow \lambda \rightarrow \mathbf{a}_i$: actionable. A mission that finds more planets must spend more time characterizing them, so the survey that maximizes yield depends on $\eta_{\oplus}$. This is why yield is sub-linear in $\eta_{\oplus}$ and why the redefined yield cannot be obtained by rescaling.

- $z_i \rightarrow C_i \rightarrow \mathbf{a}_i$: actionable. Exozodi is measured on the first visit and the survey adapts (Stark et al. 2015).
- A_{ij} , N_i , $\mathbf{o}_{ij}$: static. The planner cannot know which stars host planets, where they are, or how dark they are.

Every observational bias in this problem — the albedo penalty, the exozodi penalty, the difference between Stark’s red and purple curves — is a statement about which of these paths exist. Getting a path wrong changes a penalty by a factor of several, as Appendix A records.

3 The nodes, specified

3.1 Stars: HPIC

The target list is the HWO Preliminary Input Catalog (Tuchow et al. 2024), the list Stark et al. (2024) state they use: 11,757 dwarfs within 50 pc, 7,767 of them FGK. Stellar photon fluxes come from a blackbody at the catalogued T_{eff} and R_* ; against HPIC’s measured V for 674 stars the median residual is +0.019 mag over 5300–7300 K, and the same fluxes agree with EXOSIMS’s Pickles templates to 0.979. The survey pool is the 1,500 stars with the widest habitable zones in λ/D after a box-aware noise-floor screen.

3.2 Occurrence: $\eta_{\oplus}$, and why its width is read as one sigma

Stark et al. (2024) Sec. 3.5 derive $\eta_{\oplus}$ by integrating the Bryson et al. (2021) Kepler DR25 posteriors over their canonical domain, drawing equally from Bryson’s two bounding completeness-extrapolation cases, and quote $0.26^{+0.29}_{-0.14}$ as a mean with an 86% interval. Bryson’s posterior chains are not published: the paper’s repository omits the output directories that hold them, and the author’s site holds only the completeness contours. The distribution must therefore be reconstructed, and the choice is decided by Stark’s own Fig. 10.

Holding this model’s yield-versus- $\eta_{\oplus}$ curve at Stark’s fixed- $\eta_{\oplus}$ level (his red curve, mean 17.35), the lognormal $\eta_{\oplus}$ that turns his red curve into his purple curve has median 0.26 and ln-width 0.80, reproducing the purple curve to a total-variation distance of 0.019. Reading the quoted interval as 86%, as the text labels it, gives a ln-width of 0.52, a distance of 0.142, and a peak at 15 against the published 10. Reading it as a one-sigma (68%) interval gives ln-width 0.76, a distance of 0.024, and a peak at 12. Bryson’s Fig. 14, digitized from its vector source, shows each extrapolation case with a ln-width of about 0.8, consistent with the figure and not with the 86% reading. This model therefore reads the interval at $z = 1$ (Table 2), giving a mean of 0.343; the text’s reading remains available as a command-line option.

The redefined box enters only through $\rho_{\mathcal{B}}$, the ratio of SAG13 integrals, which does not depend on the normalization Γ . Its posterior comes from an `emcee` chain over $(\ln \Gamma, \alpha, \beta)$ with SAG13 priors, and is narrow because Γ cancels.

3.3 Exozodi: the HOSTS distribution and four measured stars

Each star’s exozodi level is drawn from the LBTI HOSTS maximum-likelihood distribution that Stark et al. (2024) use, digitized from the vector content stream of their Fig. 9 (red): a median of 2.98 zodis, against Stark’s stated three, with 21% of draws above 100 zodis. The histogram carries 0.948 of the implied $10^4 \times 500$ draws; the shortfall is renormalized rather than placed off-axis, because doing so is what reproduces the stated median. ϵ Eri (297 zodis), θ Boo (148), 72 Her (588; “w Her” in HPIC) and 110 Her (235) are pinned to their LBTI measurements, as Stark does, whenever levels are drawn.

The heavy tail matters more than the median. A star at hundreds of zodis is effectively removed from the survey; the lognormal stand-in this replaced put 0.2% of stars above 100 zodis and gave an exozodi penalty of 0.7%, against Stark’s 11%. The exozodi surface brightness follows Stark (2014) App. C: a constant habitable-zone optical depth, evaluated at the EEID, with solar colours.

3.4 Radiometry and exposure times: τ_{ij}

Exposure times follow Stark et al. (2019) Eq. 1,

$$\tau = (S/N)^2 \frac{CR_p + 2 CR_b}{CR_p^2}, \quad (4)$$

with backgrounds from leaked starlight, local zodi, exozodi and the detector (dark current and clock-induced charge with the 6.73 Geiger factor of Stark 2019 Eq. 9). The zodiacal and exozodiacal backgrounds carry the extended-source throughput $T_{\text{sky}}(r)$ of Stark (2019) Eqs. 5–6, reconstructed as $T_{\text{sky}}(r) = 0.678 \Upsilon_c(r)/\Upsilon_{c,\text{max}}$, where $0.678 = \Upsilon_{c,\text{max}}/\text{EE}(0.7 \lambda/D)$ is its large-separation value (Sec. 3.6). The coronagraph is the DMVC6 of Stark et al. (2024), digitized from the vector content stream of their coronagraph figure, evaluated in *circumscribed* λ/D (the figure’s convention) with a collecting area normalized to the full obscured primary. Detection combines the 450 nm and 550 nm channels in quadrature at S/N 7 with a $\Delta\text{mag} = 26.5$ noise floor. Characterization is an $R = 140$ IFS spectrum at S/N per spectral bin from Stark’s Fig. 3, choosing the best of nine bandpasses, with 96 pixels per spectral bin at $1 \mu\text{m}$, taken at the planet’s most favourable orbital phase.

3.5 Completeness, counting, and the allocation

For each star, 1 200–3 000 planets are injected over the box and propagated through the radiometry at up to six independent visit epochs. $C_i(\tau, k)$ is the fraction of injected planets detected within exposure τ on at least one of k visits *and* characterizable inside two months: Stark et al. (2019) state that planets failing either limit “did not count toward the yield”. The survey then maximizes $\sum_i \eta_{\mathcal{B}} C_i$ subject to a two-year budget, charging each star $k(\tau'\tau + t_{\text{oh}})$ for detection plus $\eta_{\mathcal{B}} C_i (\tau'\tau_{\text{char}} + t_{\text{oh}})$ for characterizing what it finds. Pooling every (k, τ) option onto one concave envelope and sweeping the equal-slope multiplier λ reproduces an exact dynamic-programming optimum to 0.00%.

3.6 The calibration κ , and what it was absorbing

The one input the papers do not publish is the pair of simulated two-dimensional coronagraph maps $\Upsilon_c(x, y)$ and $I(x, y)$; the digitized azimuthal averages stand in for them. A single throughput factor κ absorbs the residual, fitted at $\eta_{\oplus} = 0.24$ with exozodi *fixed* at three zodis, because Stark’s 22.5 comes from a fixed-exozodi run that precedes his exozodi sampling.

Until this revision κ was 1.62, inside the factor-of-two plausibility gate but far from unity, and the open problem below showed it was compensating for something. Stark (2019) Eqs. 5–6 write both sky backgrounds with $T_{\text{sky}}(x, y)$, “the instrument’s throughput for extended sources”, computed “by first convolving the spatially-dependent PSF at all locations with a normalized uniform background”. That is a map: background light near the inner working angle is attenuated by the same focal-plane mask that attenuates a planet there. This model had used its large-separation value, 0.678, everywhere, charging every planet near the IWA for background the mask removes. Reconstructing the map as the core-throughput curve rescaled to that value (which assumes the fraction of an off-axis PSF inside the photometric core does not change with separation, and omits the $\sim \lambda/D$ smoothing of the convolution) gives:

	T_{sky} constant	$T_{\text{sky}}(r)$
Planning yield at 6 m with no calibration ($\kappa = 1$), published 22.5	18.65	22.61
Fitted κ	1.623	0.987
Mean characterization time, first 18 EECs, 6 m / 9 m (d); published 22 / 3.5	12.6 / 1.12	13.1 / 1.30

(one exozodi seed, 1 200 planets per star; `explorations/whatIfCharacterizationTreatment.py`).

The factor was absorbing the constant T_{sky} . With the map, the uncalibrated model is 22.61 against 22.5 in the pipeline (+0.5%), and the fitted factor is $\kappa = 0.988$. This is a single number, and this project has been caught by single numbers before, so it is worth stating what it does not fix: the characterization times are essentially unchanged, because the old κ was speeding up every exposure and the map speeds up only those near the IWA, and the two roughly cancel for the easiest targets. A split calibration, one factor for detection and one for spectra, cannot repair them either: a scalar on spectra rescales the 6 m and 9 m times alike and leaves their ratio, which is the discrepancy (Sec. 8), where it is.

4 Validation against the published record

Four separate times in this project an apparent agreement rested on two errors that cancelled, so no single number is treated as evidence. Step A09 compares every reachable quantity with the published record, tags each check as independent or tuned, and records its source and tolerance. Tuned checks (the digitized coronagraph round-trip and the calibrated yield) are excluded from Table 3.

Check	Published	Model	Diff.	Tol.	Verdict
eta.Earth over the canonical box (SAG13)	0.24	0.2404	0%	3%	agrees
Uncalibrated 6 m yield	22.5	22.61	1%	20%	agrees
Calibration throughput factor	1	0.9876	1%	100%	agrees
Sampling distribution sigma (Fig. 4)	5	4.737	5%	25%	agrees
Albedo penalty on expected yield	0.12	0.08435	30%	50%	agrees
P25 including sigma_eta, 6 m	0.32	0.3572	12%	25%	agrees
Yield-aperture exponent	1.9	1.762	7%	15%	agrees
Exozodi-sampling penalty on expected yield	0.111	0.1124	1%	50%	agrees
P25 excluding sigma_eta, 6 m	0.06	0.07778	30%	50%	agrees
Mean characterization time, first 18 EECs, 6 m (days)	22	13.82	37%	30%	disagrees
Mean characterization time, first 18 EECs, 9 m (days)	3.5	1.849	47%	30%	disagrees
Characterization-time ratio 6 m / 9 m	6.2	7.475	21%	20%	disagrees
P25 including sigma_eta at 6 m	0.32	0.3585	12%	25%	agrees
P25 including sigma_eta at 7 m	0.53	0.5383	2%	25%	agrees
P25 including sigma_eta at 8 m	0.67	0.6686	0%	25%	agrees
P25 including sigma_eta at 9 m	0.78	0.7665	2%	25%	agrees
Fixed-eta yield mean (Fig. 10 red)	17.35	18.3	5%	10%	agrees
Realized-yield distribution median	18	19	6%	15%	agrees
Realized-yield fraction below 12 EECs	0.2907	0.2469	15%	35%	agrees
Realized-yield distribution mean	21	22.47	7%	30%	agrees
Realized-yield distribution mode	10	14	40%	25%	disagrees
Maximum luminosity among reachable targets	20	15.34	23%	60%	agrees
Maximum distance among reachable targets	25	23.73	5%	60%	agrees

Table 3: The 23 independent checks of step A09, of which 19 agree. Published values from Fig. 10 are digitized from the figure’s vector content stream; the fixed- η mean is Stark’s red curve, the median, mode and low tail are his purple curve.

4.1 The realized-yield distribution

Stark’s Fig. 10 purple curve (all astrophysical uncertainties, including $\sigma_{\eta_{\oplus}}$) has its peak at 10, median 18 and mean 21.0; the red curve ($\eta_{\oplus}$ fixed) has mean 17.35. The digitization reproduces three numbers stated in his text — the red mean of 17.3, and P_{25} of 6% and $\sim 32\%$ — so the published curves are known to the width of a line. This model’s realized distribution has median 19, 0.25 of its mass below 12 candidates (published 0.29), a peak at 14, and an $\eta_{\oplus}$ -fixed mean of 18.30. The two are drawn side by side in Fig. 5.

4.2 The aperture scaling

With κ frozen at its 6 m value, P_{25} including $\sigma_{\eta_{\oplus}}$ at 6/7/8/9 m is 36 / 54 / 67 / 77% against a published 32 / 53 / 67 / 78%, and the expected yield at $\eta = 0.24$ runs 18.4 / 24.8 / 31.1 / 37.5, a scaling of $D^{1.76}$ against a published $\sim D^{1.9}$. These are out-of-sample predictions: nothing was fitted at 7, 8 or 9 m.

4.3 Independent implementation: EXOSIMS

EXOSIMS shares no code or scheduling method with this pipeline. Fed identical FGK stars and optics, its count rates agree to 1.001 (local zodi), 0.975 (exozodi) and 0.979 (stellar). Reading it also exposed two missing background terms, each then confirmed against Stark’s papers. It cannot be run end to end here, because of version incompatibilities that also break its own shipped sample script.

5 Side by side with the published figures

Nine of the 26 figures in Stark et al. (2024) show a quantity this model computes under the same assumptions. Each is reproduced below next to the original, which is copied unaltered from the arXiv e-print source (arXiv:2405.19418, `ms_v2`). The model versions are drawn by step A12 in the published axes, ranges and line styles; nothing is rescaled to improve agreement. Figures without a like-for-like counterpart (optical layouts, throughput and QE curves, the design-change scenarios of his Sec. 5) are omitted.

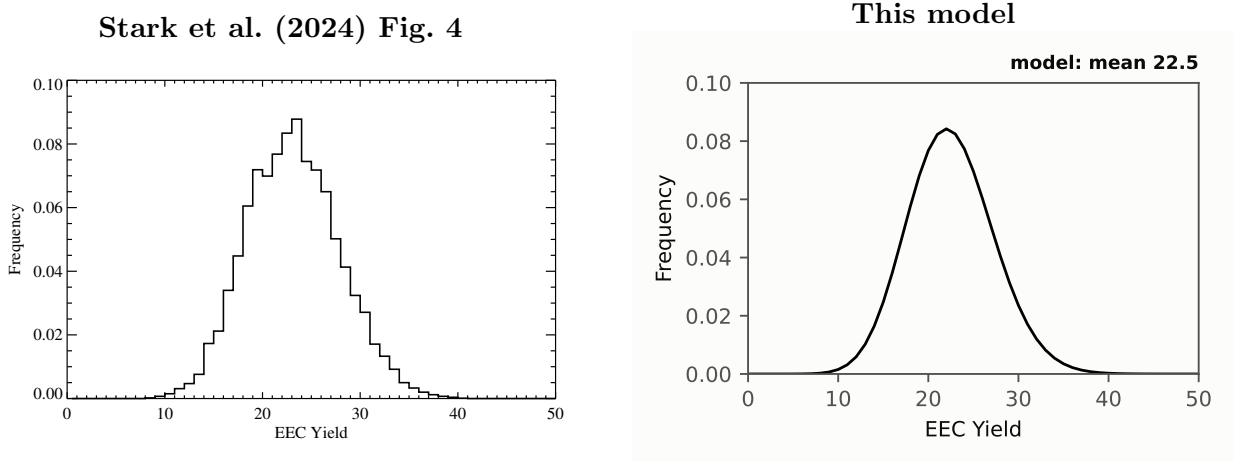
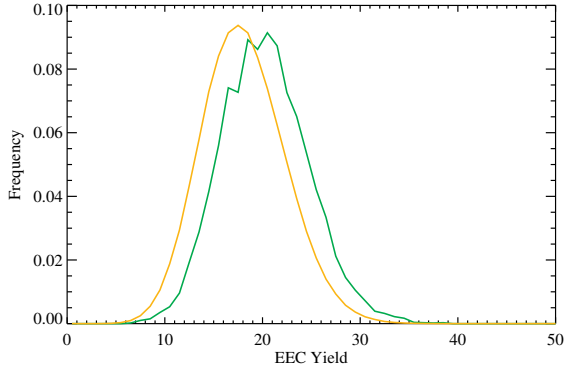


Figure 2: Exoplanet sampling only: $A_G = 0.2$, exozodi fixed at 3 zodi, Poisson counting. Published mean 22.5, standard deviation 21% of the mean. Model mean 22.5, which is *tuned*: κ is fitted to 22.5.

Stark et al. (2024) Fig. 7



This model

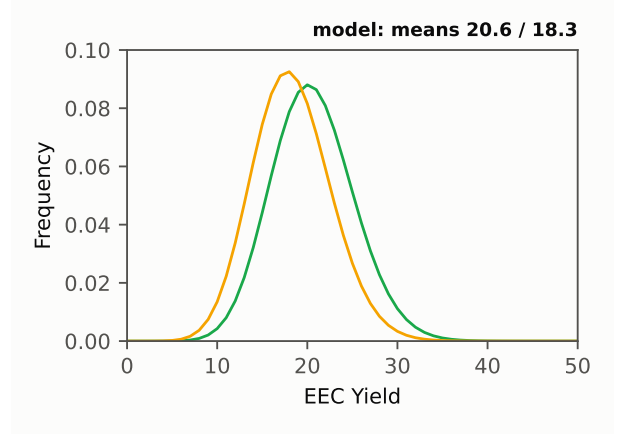
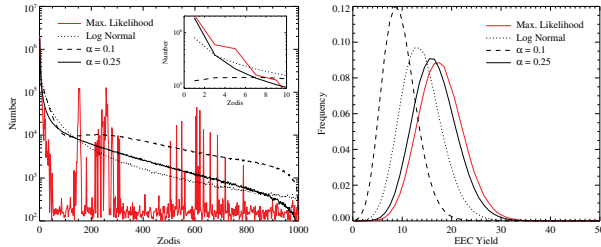


Figure 3: Albedo drawn with exozodi fixed (green), and exozodi drawn as well (orange), $\eta_{\oplus}$ fixed. Published means 19.8 and 17.6. Model means 20.6 and 18.3 over 200 draws.

Stark et al. (2024) Fig. 9



This model

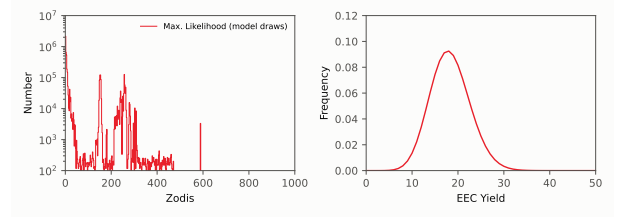
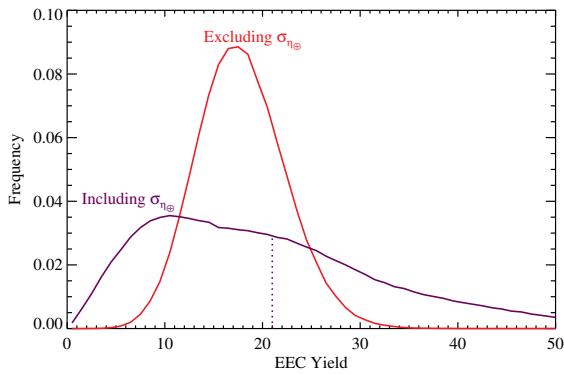


Figure 4: Left: exozodi levels drawn from the LBTI HOSTS maximum-likelihood distribution (red); the model panel is the histogram of the levels the pipeline actually drew, scaled to Stark's 500×10^4 draws (median 3.00 zodis; Stark's other three distributions are not modelled). Right: the $\eta_{\oplus}$ -fixed yield those levels produce (red).

Stark et al. (2024) Fig. 10



This model

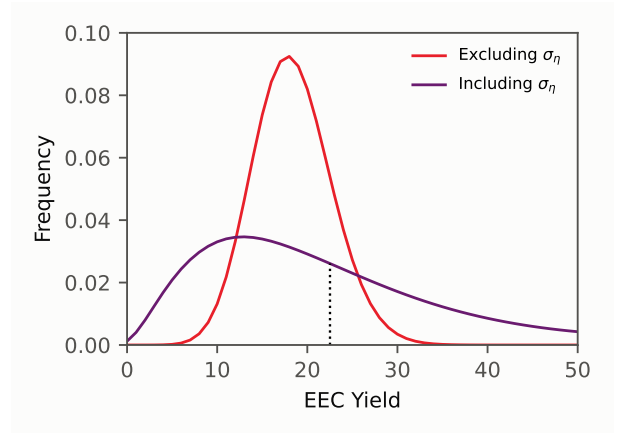
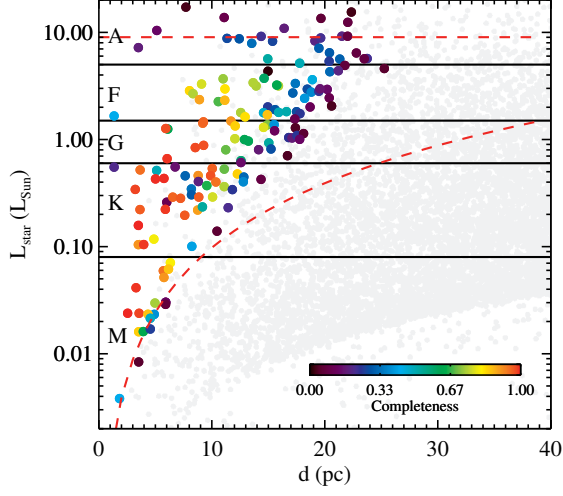


Figure 5: Realized yield including (purple) and excluding (red) $\eta_{\oplus}$ uncertainty; the dotted line marks the purple mean. Total-variation distance, model to published: 0.034 (purple), 0.097 (red). Means: published 21.0 / 17.3, model 22.5 / 18.4.

Stark et al. (2024) Fig. 11



This model

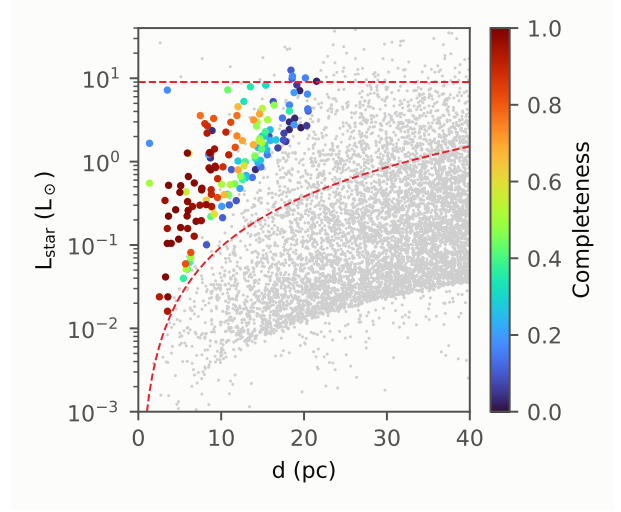
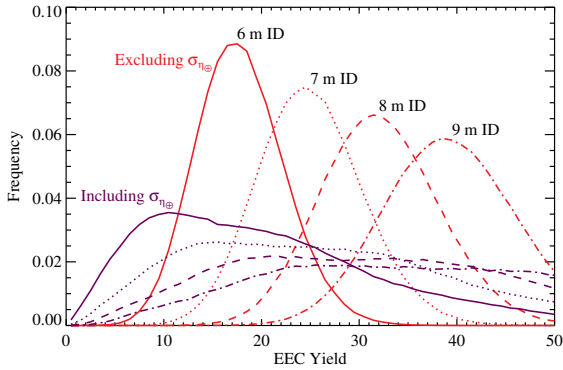


Figure 6: Targets selected in one representative exozodi draw, coloured by completeness, over the input list in grey. Published: 154 targets, summed completeness 77.4, median 0.50. Model (draw 99, the median-yield draw): 157 targets, summed 83.5, median 0.24 over the 141 matched targets. Red dashed lines: the noise floor for a $1.4 R_{\oplus}$ planet at the EEID, and the HZ outer edge at $1.5 \lambda/D$ at $1 \mu\text{m}$; the spectral-type lines are not redrawn.

Stark et al. (2024) Fig. 12



This model

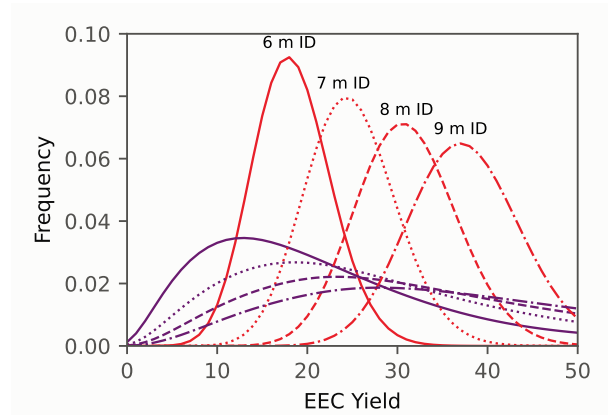
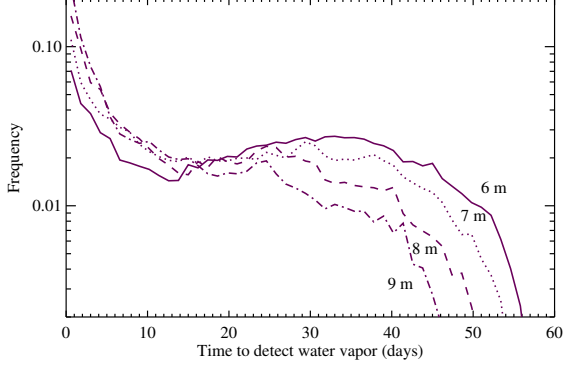


Figure 7: Realized yield at 6, 7, 8 and 9 m inscribed diameter (solid, dotted, dashed, dot-dashed), including (purple) and excluding (red) $\sigma_{\eta_{\oplus}}$, with the calibration frozen at 6 m. Published: excluding $\sigma_{\eta_{\oplus}}$, 9 m gives 2.2 times the 6 m yield; model 2.04 times.

Stark et al. (2024) Fig. 14



This model

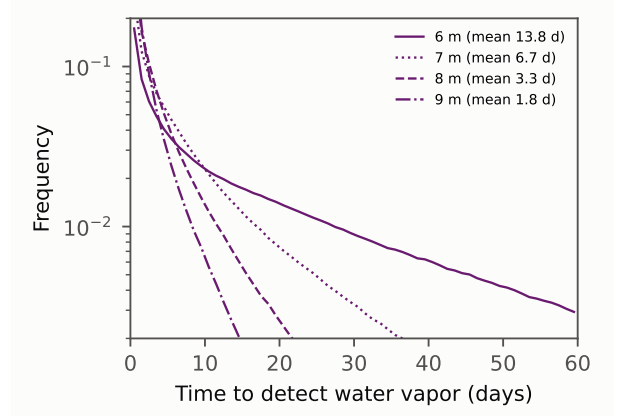
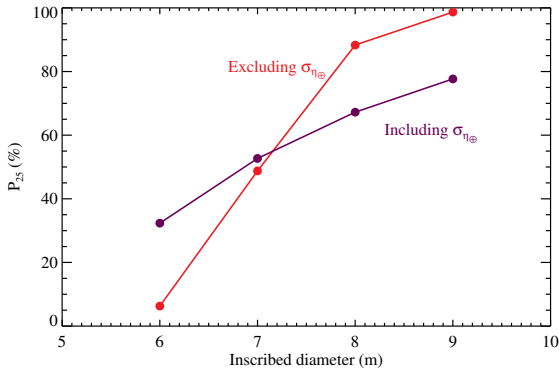


Figure 8: Characterization times of individual EECs among the first 18 at 6–9 m (model: 20 exozodi draws re-derived planet by planet, 1-day bins). Published means 22 and 3.5 days at 6 and 9 m; model means at 6/7/8/9 m: 13.8 / 6.7 / 3.3 / 1.8 days. Stark does not state his normalization, and with 1-day bins his 9 m plateau alone would exceed his stated mean, so compare shapes, not heights: his distributions keep a plateau of 15–50-day spectra at every aperture, which the model’s lack.

Stark et al. (2024) Fig. 15



This model

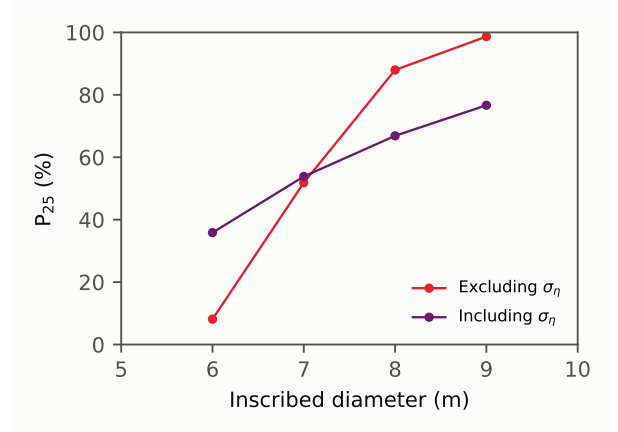
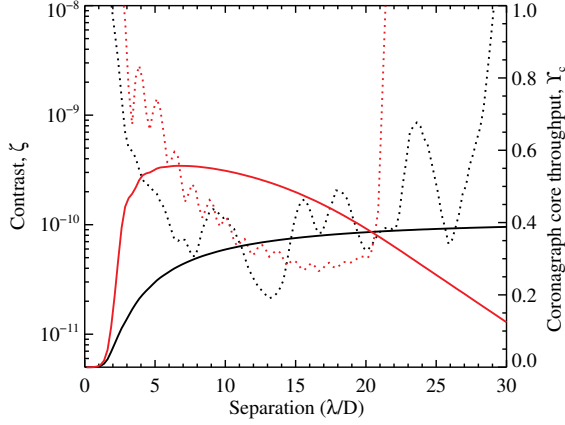


Figure 9: P_{25} against inscribed diameter. Published including $\sigma_{\eta_{\oplus}}$ 32/53/67/78%, model 36 / 54 / 67 / 77%; excluding, published 6/49/89/99.5%, model 8 / 52 / 88 / 99%.

Stark et al. (2024) Fig. 25



This model

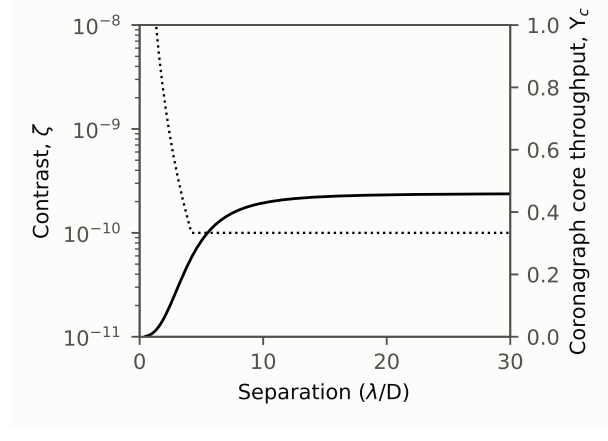


Figure 10: DMVC6 raw contrast (dotted) and core throughput (solid) against separation in circumscribed λ/D . The model curves are the digitized published ones, so this pair checks the digitization, not the model. The model's contrast is shown with the 10^{-10} floor applied, as the exposure times use it; the published curve is the raw simulation. The published figure also shows the PIAA-FPM2.5 (red), which is not used.

6 Results

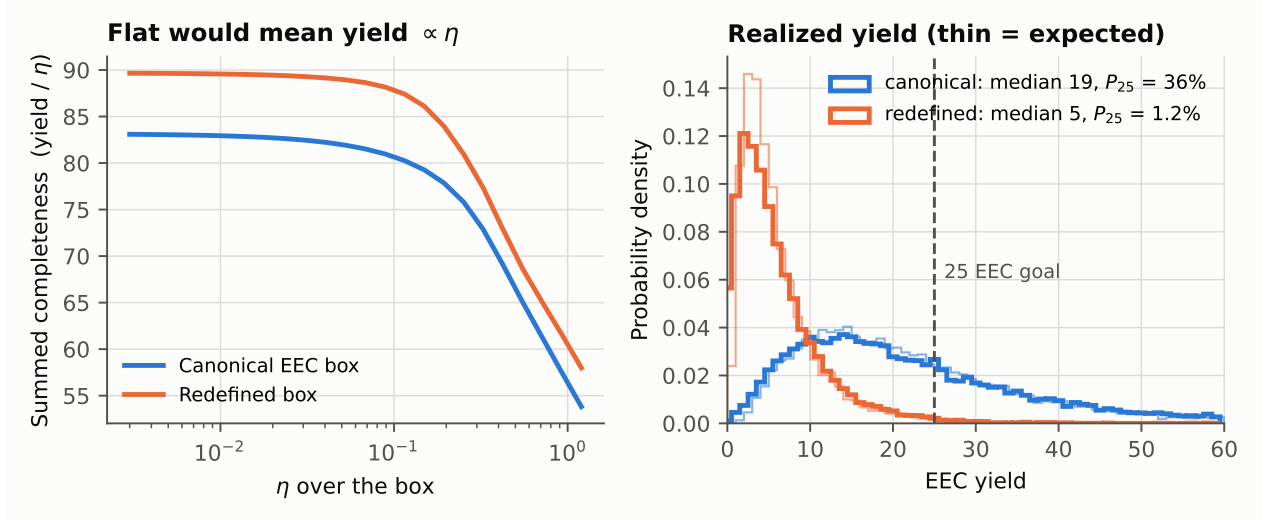


Figure 11: Left: summed completeness against η ; its decline is the characterization burden consuming search time, i.e. the $\eta_{\oplus} \rightarrow \lambda$ edge of Fig. 1. Right: the realized yield distributions, Eq. (3) marginalized over $\eta_{\oplus}$ (heavy), and the expected yield Λ alone (thin).

6.1 Why the yield ratio exceeds the occurrence ratio

The occurrence ratio 0.204 decomposes into about 0.32 from narrowing the habitable zone and 0.63 from the mass selection. The yield ratio, 0.245, is larger, for three reasons that are all visible in Fig. 1:

1. **Freed characterization time** ($\eta_{\mathcal{B}} \rightarrow \lambda$). Fewer expected detections leave more of the budget for searching. Raises the ratio.
2. **The faintest planets are removed** ($\mathcal{B} \rightarrow \mathbf{o}_{ij}$). Truncating at 1.20 scaled AU removes planets up to $(1.67/1.20)^2 \approx 1.9$ times fainter than those kept. Raises the ratio.

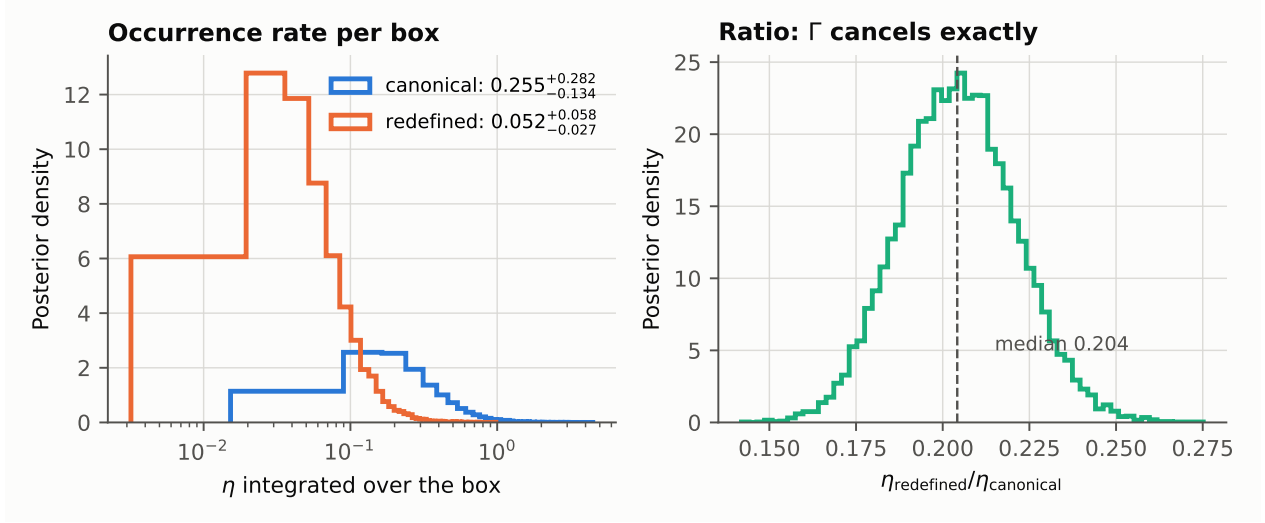


Figure 12: Left: the occurrence rate integrated over each box. Right: their ratio ρ_B , tight because Γ cancels.

3. **Distant stars are lost** ($\mathcal{B} \rightarrow C_i$). For distant stars only the outer habitable zone clears the inner working angle. Lowers the ratio.

The first two win. Under the redefined selection a 6 m telescope reaches 25 candidates with probability 1.19%: a narrow habitable zone and a mass-selected sample put the decadal benchmark out of reach at this aperture for any plausible $\eta_{\oplus}$.

7 The Earth zone and the classic habitable zone, star by star

The *Earth zone* is this report’s redefined box: 0.96–1.20 AU for the Sun, scaled by $\sqrt{L_{\star}}$, holding planets of $0.5\text{--}2 M_{\oplus}$. The *classic habitable zone* is Stark’s canonical box, 0.95–1.67 AU scaled the same way, holding planets up to $1.4 R_{\oplus}$. Step A11 tabulates both for every one of the 1,500 screened stars, in AU, in milliarcseconds and in λ/D of the 6 m telescope at 550 and 1000 nm, together with what each zone’s own optimized survey expects to find there: each box’s survey is re-optimized in every exozodi draw at its posterior-median occurrence rate ($\eta = 0.255$ and 0.0520), and a star’s expected yield is η times its completeness averaged over draws. The full table is `TabulateEarthZones/earthZoneTable.csv`; Appendix B lists the 183 stars that contribute in either survey.

Totals. The Earth zone is a third as wide as the classic zone in scaled AU and holds 0.204 as many candidates per star. Summed over stars the expected yield is 19.30 in the classic zone and 4.63 in the Earth zone, a ratio of 0.240; star by star the ratio has median 0.241 (68% of stars between 0.201 and 0.325). The classic survey observes 184 stars in at least half the draws at a mean completeness of 0.45; the Earth-zone survey observes 171 at 0.54. The narrower zone is easier to complete, because its planets sit closer in and are brighter, so the survey concentrates on slightly fewer stars and reaches a larger fraction of their candidates. The ratio rises from about 0.21 for the faintest, nearest stars to about 0.33 for stars above $5 L_{\odot}$ (Table 4): around distant and luminous stars the classic zone’s outer planets are too faint to be found, so losing them costs little.

Angle. The Earth zone sits at the inner edge of the classic zone, so in angle it is the harder part of it. With the half-throughput inner working angle at $2.93 \lambda/D$ (inscribed), a mean 91% of the Earth zone lies outside it at 550 nm against 96% of the classic zone, and at 1000 nm, where water is sought, 47% against 64%, averaged over the stars either survey observes. The coronagraph still passes light inside that angle, which is why the loss in yield is smaller than the loss in angle.

Luminosity ($L_{\odot}$)	stars	stars observed		expected EECs		ratio
		classic	Earth	classic	Earth	
$L < 0.3$	47	36	33	5.57	1.15	0.21
0.3-0.6	75	30	26	4.11	0.91	0.22
0.6-1	102	19	17	1.95	0.44	0.23
1-2	294	33	30	3.13	0.78	0.25
2-5	665	44	42	3.72	1.07	0.29
$L > 5$	317	22	23	0.82	0.27	0.33
Distance						
< 5 pc	15	15	15	2.97	0.66	0.22
5-10 pc	59	59	57	9.60	2.16	0.23
10-15 pc	95	65	58	5.16	1.37	0.27
15-20 pc	133	42	37	1.50	0.42	0.28
> 20 pc	1198	3	4	0.08	0.02	0.21

Table 4: Where each zone’s yield comes from. “Observed” means allocated time in at least half of the exozodi draws. Each zone’s survey is optimized for its own candidates.

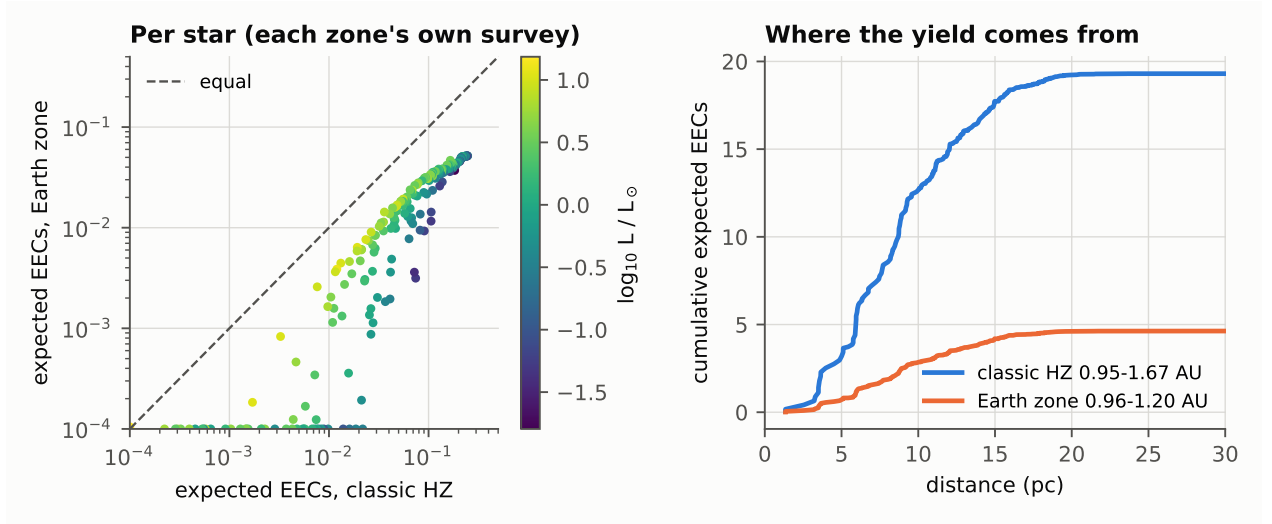


Figure 13: Left: each star’s expected Earth-zone yield against its classic-zone yield, coloured by luminosity. Right: cumulative expected yield with distance for both zones.

8 The open problem: characterization time and the target mix

With κ explained and exozodi marginalized, the level problem of earlier revisions is gone: the fixed- $\eta_{\oplus}$ yield is 18.30 against Stark’s 17.35 (5.5% high, inside the 10% the researcher set as the target), the exozodi penalty is 11.2% against 11.1%, and P_{25} agrees at every aperture. What remains is one discrepancy seen three ways.

The symptom. The mean characterization time of the first 18 EECs is 13.8 d at 6 m and 1.85 d at 9 m, against Stark’s 22 and 3.5 (Sec. 4.1): the easiest targets are characterized too fast, by a factor of about two. The shape says where the difference is (Fig. 8): Stark’s first 18 EECs include a plateau of 15–50-day spectra at every aperture, while this model’s individual characterization times fall away steeply, with almost none above about 15 days at 9 m. The peak of the realized distribution sits at 14 rather than 10.

The target mix differs. Against Stark’s Fig. 11 (Fig. 6), in the same characterization-gated completeness and one representative exozodi draw, this model matches the nearest stars — median 0.98 against 0.95 inside 6 pc — but gives stars at 10–15 pc a median of 0.35 (published 0.73) and at 15–20 pc 0.07 (0.28). By luminosity, stars of 1–2 $L_{\odot}$ get 0.08 (published 0.62) and stars of 2–5 $L_{\odot}$ get 0.14 (0.47), with 42% of the latter allocated essentially nothing. The survey concentrates on nearby stars instead, and reaches Stark’s total by completing them more fully.

The cause is characterization cost. At $R = 140$ near 1 μm the characterization background is exozodi-dominated, and the exozodi count rate is a surface brightness, nearly the same for every star, while the planet dims as d^{-2} ; characterization time therefore grows roughly as d^4 . For an Earth twin at quadrature the exozodi-to-planet count ratio per spectral bin is 1.3 at τ Cet, 7 at 61 Vir, 20 at θ Boo and 36 at 110 Her, and the characterization times are 0.21, 10, 72 and 255 days against a 60-day cap. Distant, luminous targets are priced out of the survey while nearby ones are characterized in hours to days.

Ruled out, each by direct test.

- *The noise floor:* removing it entirely leaves luminous-star completeness unchanged.
- *The detector saturation definition:* Stark (2019) sets CR_{sat} to ten times an Earth twin’s core-pixel rate, where this model uses the scene; characterization times change by under 1%.
- *IFS pixels:* 96 per spectral bin matches Stark (2024) Table 2. Latouf et al. (2026) use 6 for a different instrument.
- *Detector noise generally:* Stark’s Scenario D (Skipper CCD) gains 1.5 $\times$ in characterization time, which his text attributes mostly to a 30% throughput gain; the implied detector share, about 12%, matches this model’s.
- *The first-18 ordering:* ordering targets by survey priority rather than by characterization cost changes the mean by 1%.
- *The Fig. 11 metric:* detection-only completeness fixes the low-luminosity stars but not the luminous ones.
- *Dust colour:* the zero point is a solar-coloured source normalized at V , which is the treatment Stark (2015) describes.

Not the calibration. Before this revision $\kappa = 1.62$ multiplied spectra as well as detections, and the obvious suspicion was that it sped up the easy spectra. It did, but removing it through the T_{sky} map left the first-18 times unchanged (Sec. 3.6), and no single factor on spectra can change their 6 m/9 m ratio, 7.5 here against 6.2.

Reading. The model’s characterization time rises with target difficulty more steeply than Stark’s: nearby targets are cheaper and distant ones dearer. Both ends move the same way under the

exozodi-limited d^4 scaling above, which this model follows closely; Stark’s numbers behave as if something flattens that dependence. Detector noise, the obvious candidate, is argued against in the list above; what remains untested includes how Stark schedules a spectrum against the planet’s orbit (this model takes the most favourable of 40 phases, which is the cheapest possible choice), whether his S/N per spectral bin counts one bin or a band average, and the exozodi surface brightness at the planet’s own separation: Stark (2019) Table 1 notes that it “varies with spectral type and planet-star separation”, while this model evaluates it at the EEID. Those are the next things to test. By the standard this project has held to, a fix must move the first-18 times at both apertures, the distant targets of Fig. 11 and the peak of Fig. 10 together.

9 Limitations

- **$T_{\text{sky}}(r)$ is reconstructed.** Stark’s map comes from 2-D PSF simulations that are not published; this model scales the azimuthally averaged core-throughput curve, which assumes a separation-independent core fraction and omits the convolution’s smoothing. It removed the need for κ (now 0.988), but a different reconstruction would move the absolute yields.
- **The coronagraph is azimuthally averaged.** The digitized one-dimensional DMVC6 curves stand in for two-dimensional maps.
- **Exozodi draws are interpolated.** 200 draws (Stark: 500) evaluated on a 23-level grid; interpolation changes optimized yields by at most 0.24%. Stark also draws the HOSTS distribution itself from its posterior, which he finds negligible and which is not modelled.
- **The $\eta_{\oplus}$ distribution is reconstructed.** Its width is inferred from Stark’s Fig. 10 because the Bryson chains are unpublished; the text’s 86% reading is one option away.
- **α, β are prior-dominated.** The ratio ρ_B inherits SAG13’s shape, not a re-fit to Kepler DR25.
- **The area fill factor (0.785) is unsourced.** It moves the yield by about $\pm 4\%$.

10 Reproducing this

The project is a twelve-step vaibify pipeline in `/workspace/yields`. Every step declares its inputs and outputs and carries integrity, qualitative and quantitative tests generated from its outputs; the physics library carries 78 unit tests. Every number in this report is injected from a step or exploration output at build time (`cd doc && python3 buildReport.py`), so a re-run cannot leave a stale claim in print.

A Ledger of hypotheses

Every proposed explanation for a disagreement with the published record, with its outcome. A defect can be genuine, worth fixing, and irrelevant to the symptom that led to it, so “real error, not the cause” is kept separate from “the cause”.

Step	Name	Nodes of Fig. 1 it implements
A01	buildTargetCatalog	$\mathbf{s}_i$: HPIC, 11,757 stars
A02	modelCoronagraph	$\mathcal{M}, \mathcal{B}, p_z$ (digitized Fig. 9), pinned z_i, p_A
A03	calibrateToStark	κ , at fixed exozodi
A04	computeCompleteness	$\mathbf{o}_{ij}, A_{ij}, \boldsymbol{\tau}_{ij}, C_i$ on a grid of z ; 200 draws of $\mathbf{z}$
A05	optimizeSurvey	$\lambda, \mathbf{a}_i$ at $\eta = 0.24$, at fixed z and per draw
A06	sampleOccurrencePosterior	$\eta_{\oplus}, \alpha, \beta, \rho_{\mathcal{B}}$
A07	predictRedefinedYield	$\mathbf{a}_i(\eta)$ on a grid, Y by Eq. (3)
A08	compareApertureScaling	the same at 6–9 m with κ frozen, over draws of $\mathbf{z}$
A09	verifyAgainstPublished	Table 3
A10	compareTargetCompleteness	per-target comparison with Fig. 11
A11	tabulateEarthZones	Sec. 7, Appendix B
A12	comparePublishedFigures	Sec. 5

Table 5: Pipeline steps mapped onto the graphical model.

Hypothesis	Outcome	Evidence
Missing revisits	Real, a cause	Single-visit only; modelling up to six visits raised the uncalibrated yield $1.7\times$.
Characterization-time statistic	Real, a cause	A median over all detectable planets, where the budget needs a mean over those counted; $D^{1.11} \rightarrow D^{1.88}$.
Over-cap characterization charged	Real, a cause	15 000 days charged at 20 pc for an observation never attempted.
Reconstructed target catalog	Real, a cause	HPIC was reachable; the Gaia stand-in lacked 26–34% of distant FGK dwarfs and ran hot.
Coronagraph aperture convention	Real, the cause	Curves are in circumscribed λ/D ; the IWA sat 19% too far out. Yield $14.7 \rightarrow 18.7$.
Collecting area as inscribed circle	Real, partial	Charged the Lyot discard twice.
Characterization at detection phase	Real, a cause	Stark (2019) Sec. 5: the orbit is known, so “the phase of the planet could be optimized.”
Characterization budget under albedo draw	Real, a cause	Stark (2024) Sec. 3.2 fixes it at $A_G = 0.2$; albedo penalty $0.050 \rightarrow 0.087$.
Constant T_{sky}	Real, the cause of κ	Stark (2019) Eqs. 5–6 make it a map; with it the uncalibrated yield is 22.61 and κ falls $1.62 \rightarrow 0.988$. Characterization times barely move.
One exozodi draw	Real, fixed	Now 200 draws with the survey re-optimized per draw; exozodi penalty 11.2% (Stark 11.1%).
Split κ (detection vs spectra)	Rejected	A scalar on spectra cannot change the 6/9 m characterization-time ratio.
Exozodi drawn during calibration	Real, a cause	κ fitted to a drawn-exozodi yield cancels the exozodi penalty.
Lognormal exozodi stand-in	Real, a cause	0.2% of stars above 100 zodis vs 21% in HOSTS; penalty $0.7\% \rightarrow 11.2\%$.
Pinned LBTI stars omitted	Real, partial	Worth at most 0.47 EEC here.
$\eta_{\oplus}$ interval read as 86%	Real, the cause of the shape	Fig. 10 requires ln-width ~ 0.8 ; distance $0.142 \rightarrow 0.024$.
Published mode ill-determined	Rejected	Stark’s 498 runs carry 1000 planet draws each; an earlier bootstrap resampled only 498 yields.
Fig. 4 mean compared with the wrong yield	Real, not the cause	Fig. 4 is sampling only at $A_G = 0.2$; the check is now marked tuned.
T_{sky} missing; exozodi law	Real, not the cause	Backgrounds $1.48\times$ too bright; constant-surface-brightness law. Together 1%.
Contrast floor inside the knee	Real, not the cause	Floor applied universally.
Noise-floor screen	Real, not the cause	Admitted nothing above $4.6 L_{\odot}$.
Screening cap, uniform bias, band-pass choice	Rejected	1.3%, destroys 7–9 m, 0.01.
Two detections for orbits	Rejected	Shifts the curve, does not narrow it.
Equal-slope allocation	Rejected	0.00% from an exact optimum.
Blackbody stellar flux	Rejected	+0.019 mag against measured V .
Stark’s per-visit albedo test	Rejected	Gives a weaker penalty (3.3%).
Noise floor, CIC definition, IFS pixels, ordering, dust colour	Rejected	Sec. 8.

Table 6: The ledger. The traps that recur — compensating errors, comparing against the wrong published curve, biased diagnostic sub-samples, and read-off values — are documented in `doc/handoffToNextAgent.md`.

B Every contributing star

Stars with an expected yield of at least 0.01 EEC in either zone's survey, in order of their classic-zone yield. Zone edges are angular separations in milliarcseconds; C is the completeness reached, averaged over exozodi draws; N is the expected number of candidates. The complete table, with every screened star and every column (AU, λ/D at 550 and 1000 nm, time allocated, characterization time, fraction of draws observed), is `TabulateEarthZones/earthZoneTable.csv`.

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
tau Cet	G8V	3.65	0.52	189–237	188–330	0.98	1.00	0.247	0.052	0.21
HD 95735	M2+V	2.55	0.0239	58–73	58–101	0.98	1.00	0.245	0.052	0.21
eps Ind	K5V	3.64	0.222	124–155	123–216	0.98	1.00	0.245	0.052	0.21
61 Cyg A	K5V	3.50	0.158	109–136	108–190	0.97	1.00	0.244	0.052	0.21
61 Cyg B	K7V	3.50	0.104	89–111	88–154	0.97	1.00	0.242	0.052	0.21
HD 217987	M2V	3.29	0.0413	59–74	59–103	0.96	1.00	0.239	0.052	0.22
70 Oph A	K0V	5.11	0.514	135–168	133–234	0.95	1.00	0.235	0.051	0.22
del Equ	F7(V)+G	4.51	0.265	110–137	108–191	0.95	1.00	0.233	0.051	0.22
V* AX Mic	M1V	3.97	0.105	78–98	77–136	0.94	1.00	0.232	0.051	0.22
omi02 Eri	K0V	5.01	0.428	125–157	124–218	0.95	1.00	0.232	0.051	0.22
eps Eri	K2V	3.22	0.342	174–218	173–303	0.94	1.00	0.229	0.051	0.22
70 Oph B	K4V	5.10	0.162	76–95	75–132	0.91	0.99	0.223	0.051	0.23
e Eri	G6V	6.04	0.663	129–162	128–225	0.91	0.99	0.221	0.051	0.23
eta Cas	F9V	5.92	1.27	182–228	181–317	0.90	0.99	0.218	0.051	0.23
36 Oph	K2V+K1V	5.95	0.334	93–117	92–162	0.89	0.97	0.218	0.050	0.23
sig Dra	K0V	5.76	0.433	110–137	108–191	0.89	0.99	0.217	0.051	0.23
del Pav	G8IV	6.10	1.25	176–220	174–306	0.89	0.99	0.216	0.051	0.23
IRAS 20079-3614		6.01	0.26	81–102	81–142	0.88	0.95	0.215	0.049	0.23
HD 131977	K4V	5.89	0.223	77–96	76–134	0.87	0.95	0.212	0.049	0.23
HD 88230	K6VeFe-	4.87	0.117	68–84	67–118	0.88	0.98	0.212	0.050	0.23
36 Oph B	K1V	5.95	0.331	93–116	92–162	0.87	0.97	0.212	0.049	0.23
ksi Boo	G7Ve+K5	6.75	0.554	106–132	105–184	0.86	0.93	0.210	0.048	0.23
V* GX And	M2V	3.56	0.0239	42–52	41–72	0.85	0.89	0.206	0.046	0.22
HD 219134	K3V	6.54	0.29	79–99	78–137	0.84	0.90	0.205	0.046	0.22
V* V2215 Oph	K5V	5.95	0.158	64–80	63–111	0.83	0.89	0.202	0.045	0.22
107 Psc	K1V	7.64	0.458	85–106	84–148	0.82	0.88	0.200	0.045	0.23
bet CVn	G0V	8.47	1.22	125–156	124–218	0.79	0.86	0.193	0.044	0.23
61 Vir	G6.5V	8.53	0.841	103–129	102–179	0.79	0.84	0.191	0.043	0.23
mu. Cas	G5Vb	7.68	0.479	87–108	86–151	0.78	0.83	0.189	0.042	0.22
HD 16160	K3V	7.23	0.282	71–88	70–123	0.76	0.80	0.185	0.041	0.22
HD 173739	M3V	3.52	0.016	35–43	34–60	0.76	0.72	0.183	0.037	0.20
chi01 Ori	G0V	8.70	1.11	116–145	115–202	0.74	0.80	0.180	0.041	0.23
ksi UMa A	F8.5:V	8.73	1.26	123–154	122–214	0.74	0.81	0.179	0.041	0.23
HD 156384	K3V+K5V	6.97	0.192	60–76	60–105	0.74	0.77	0.178	0.040	0.22
chi Dra	F7V	8.30	2.19	171–214	170–298	0.73	0.85	0.175	0.043	0.25
ksi UMa	F8.5:V+	8.73	0.796	98–123	97–171	0.72	0.77	0.174	0.039	0.23
HD 4628	K2.5V	7.43	0.3	71–88	70–123	0.72	0.76	0.174	0.039	0.22
zet Tuc	F9.5V	8.61	1.28	126–158	125–220	0.71	0.77	0.173	0.039	0.23
p Eri A	K2V	8.19	0.345	69–86	68–120	0.72	0.77	0.172	0.040	0.23
V* TW PsA	K4Ve	7.60	0.196	56–70	55–97	0.72	0.79	0.172	0.041	0.24
mu.01 Her	G5IV	8.33	2.68	188–235	186–328	0.73	0.88	0.171	0.044	0.26
HD 192310	K2+V	8.81	0.405	69–87	69–121	0.72	0.79	0.171	0.041	0.24
bet Com	F9.5V	9.20	1.42	124–155	123–216	0.71	0.77	0.170	0.039	0.23
pi.03 Ori	F6V	8.02	2.85	202–253	200–352	0.73	0.89	0.170	0.044	0.26
p Eri B	K2V	8.20	0.305	65–81	64–112	0.71	0.76	0.169	0.039	0.23
gam Lep	F6V	8.90	2.36	166–207	164–288	0.72	0.81	0.169	0.041	0.24
ksi Boo B	K5Ve	6.75	0.127	51–63	50–88	0.70	0.75	0.168	0.038	0.23
gam Pav	F9VFe-1	9.26	1.46	125–157	124–218	0.70	0.77	0.168	0.039	0.23
kap01 Cet	G5V	9.27	0.881	97–122	96–169	0.70	0.77	0.167	0.039	0.24
bet Hyi	G0V	7.48	3.57	243–303	240–422	0.72	0.95	0.165	0.047	0.28
HD 102365	G2V	9.32	0.849	95–119	94–165	0.69	0.77	0.165	0.039	0.24
tet Per	F8V	11.14	2.32	131–164	130–228	0.70	0.83	0.164	0.042	0.26
HD 131976	M1.5V	5.90	0.0723	44–55	43–76	0.68	0.70	0.164	0.036	0.22
41 Ara	G9V	8.79	0.461	74–93	73–129	0.69	0.76	0.163	0.039	0.24
iot Per	G0V	10.57	2.26	136–171	135–237	0.68	0.79	0.159	0.040	0.25
61 UMa	G8V	9.57	0.625	79–99	78–138	0.67	0.77	0.158	0.039	0.25
del Tri	G0.5VFe	10.91	1.21	97–121	96–168	0.67	0.81	0.157	0.042	0.26
del Eri	K0+IV	9.09	3.29	192–240	190–333	0.66	0.85	0.153	0.042	0.27
alf Men	G7V	10.21	0.866	87–109	87–152	0.64	0.74	0.151	0.038	0.25
HD 32147	K3+V	8.84	0.29	58–73	58–102	0.60	0.69	0.141	0.035	0.25
HD 10780	K0V	10.04	0.536	70–88	69–122	0.61	0.75	0.140	0.038	0.27
12 Oph	K1V	9.89	0.46	66–82	65–114	0.60	0.73	0.138	0.037	0.27

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
lam Ser	G0-V	11.90	2.03	115–144	114–200	0.60	0.75	0.137	0.038	0.28
HD 79210	K7V	6.33	0.0813	43–54	43–75	0.58	0.56	0.137	0.029	0.21
gam Ser	F6V	11.16	2.96	148–185	147–258	0.59	0.72	0.136	0.036	0.27
iot Peg	F5V	11.90	3.57	152–191	151–265	0.59	0.78	0.134	0.038	0.28
bet Vir	F9V	11.00	3.69	168–209	166–292	0.59	0.77	0.133	0.037	0.28
V* AK Lep	K3	8.89	0.279	57–71	56–99	0.57	0.66	0.133	0.034	0.26
HD 36395	M1.5Ve	5.70	0.0594	41–51	41–71	0.54	0.50	0.129	0.026	0.20
zet Dor	F9VFe-0	11.69	1.48	100–125	99–174	0.56	0.74	0.129	0.038	0.29
alf Cen B	K1V	1.35	0.552	529–661	524–921	0.50	0.68	0.128	0.035	0.27
11 LMi	G8Va	11.23	0.801	77–96	76–133	0.56	0.72	0.127	0.037	0.29
lam Aur	G1.5IV-	12.56	1.78	102–128	101–178	0.55	0.75	0.126	0.038	0.30
zet TrA	F9V	12.07	1.34	92–115	91–160	0.54	0.73	0.123	0.037	0.30
20 Crt	K0V	9.56	0.371	61–77	61–106	0.50	0.60	0.117	0.031	0.26
ups And	F9V	13.47	3.38	131–164	130–228	0.52	0.75	0.117	0.036	0.31
36 UMa	F8V	12.94	1.63	95–118	94–165	0.49	0.71	0.110	0.036	0.32
gam Vir	F1-F2V	12.02	4.55	170–213	169–296	0.50	0.71	0.110	0.033	0.30
HD 50281	K3.5V	8.74	0.219	51–64	51–89	0.47	0.46	0.110	0.023	0.21
10 Tau	F9IV-V	13.92	3.16	123–153	121–213	0.48	0.70	0.108	0.034	0.32
zet02 Ret	G1V	12.04	1.02	80–100	80–140	0.48	0.66	0.107	0.034	0.31
HD 79211	M0V	6.33	0.0707	40–50	40–70	0.45	0.28	0.106	0.014	0.13
HD 42581	M1V	5.76	0.0515	38–47	37–66	0.45	0.23	0.106	0.012	0.11
iot Psc	F7V	13.66	3.39	129–162	128–225	0.47	0.68	0.106	0.033	0.31
alf For	F6V	14.00	4.28	142–177	140–247	0.48	0.72	0.104	0.034	0.32
zet01 Ret	G2.5VHd	12.04	0.795	71–89	70–124	0.45	0.58	0.101	0.029	0.29
gam Vir B	F0mF2V	12.74	5.27	173–216	171–301	0.45	0.70	0.099	0.031	0.31
HD 17925	K1V	10.36	0.401	59–73	58–102	0.42	0.43	0.096	0.022	0.22
tau01 Eri	F7V	14.27	2.67	110–137	109–191	0.41	0.63	0.094	0.032	0.34
psi Cap	F5V	14.63	3.72	127–158	125–220	0.42	0.65	0.092	0.031	0.34
alf Com	F5V+F6V	14.32	3	116–145	115–202	0.41	0.63	0.091	0.031	0.34
HD 191849	M0V	6.16	0.0616	39–48	38–67	0.39	0.18	0.091	0.009	0.10
54 Psc	K0.5V	11.10	0.529	63–79	62–109	0.40	0.45	0.089	0.022	0.25
47 UMa	G1-VFe-	13.88	1.6	87–109	86–152	0.40	0.58	0.086	0.029	0.34
HD 103095	K1V_Fe-	9.17	0.235	51–63	50–88	0.36	0.27	0.083	0.014	0.17
HD 157881	K7V	7.71	0.123	44–55	43–76	0.36	0.19	0.082	0.009	0.11
alf Crv	F1V	14.97	4.34	134–167	132–232	0.36	0.60	0.079	0.028	0.35
eta Lep	F2V	14.95	5.66	153–191	151–266	0.37	0.62	0.077	0.027	0.35
HD 104304	G8IV	12.76	0.892	71–89	70–124	0.35	0.41	0.077	0.021	0.27
eta Cas B	K7Ve	5.93	0.0513	37–46	36–64	0.30	0.06	0.074	0.003	0.04
chi Her	F8VFe-2	15.90	3.16	107–134	106–187	0.33	0.54	0.073	0.026	0.36
HD 147513	G5V	12.89	0.998	74–93	74–129	0.32	0.42	0.072	0.021	0.29
HD 119850	M2V	5.43	0.0397	35–44	35–61	0.31	0.08	0.072	0.004	0.05
111 Tau	F8V	14.57	1.78	88–110	87–153	0.32	0.45	0.069	0.022	0.31
HD 223778	K3V	10.89	0.399	56–70	55–97	0.31	0.22	0.069	0.011	0.16
zet Her	G0IV	10.80	7.26	240–300	237–417	0.32	0.56	0.068	0.024	0.34
V* DE Boo	K0.5V	11.63	0.541	61–76	60–106	0.30	0.25	0.068	0.013	0.19
ksi Peg	F6V	16.39	4.76	128–160	126–222	0.31	0.51	0.067	0.023	0.35
HD 13445	K1.5V	10.76	0.406	57–71	56–99	0.29	0.23	0.067	0.012	0.18
HD 10307	G1V	14.67	1.89	90–112	89–157	0.30	0.46	0.066	0.023	0.34
58 Eri	G2.5IV-	13.23	0.966	71–89	71–124	0.29	0.31	0.065	0.016	0.24
tau Boo	F7IV-V	15.60	3.18	110–137	109–191	0.30	0.49	0.065	0.024	0.36
HD 122064	K3V	10.06	0.291	51–64	51–89	0.29	0.16	0.064	0.008	0.12
85 Peg	G5VbFe-	12.32	0.674	64–80	63–111	0.28	0.26	0.062	0.013	0.21
26 Dra	F9V+K3V	14.43	1.48	81–101	80–141	0.28	0.37	0.061	0.018	0.30
sig Boo	F4VbF2m	15.75	3.22	109–137	108–190	0.28	0.41	0.060	0.020	0.33
nu. Phe	F9VFe+0	15.25	2.03	90–112	89–156	0.27	0.38	0.058	0.018	0.31
HD 211415	G0V	14.06	1.17	74–92	73–129	0.27	0.28	0.058	0.014	0.24
HD 84117	F9V	14.95	2.01	91–114	90–159	0.26	0.37	0.056	0.019	0.33
tet UMa	F7V	13.53	7.9	199–249	197–347	0.27	0.48	0.056	0.019	0.35
b Aql	G7IVHde	14.91	1.7	84–105	83–146	0.26	0.35	0.055	0.017	0.30
h Dra	F8V	15.33	2.28	95–118	94–164	0.25	0.37	0.055	0.018	0.33
m Tau	G4V	15.91	2.41	94–117	93–163	0.26	0.38	0.053	0.018	0.35
I Car	F3V	16.28	5.25	135–169	134–235	0.24	0.41	0.053	0.018	0.35
del Aql	F1IV-V(	15.37	8.26	179–224	178–312	0.24	0.42	0.048	0.017	0.35
mu. Ara	G3IV-V	15.60	1.9	85–106	84–148	0.23	0.30	0.048	0.015	0.31
eta Boo	G0IV	11.34	8.79	251–314	248–437	0.23	0.41	0.048	0.016	0.34
18 Sco	G2Va	14.13	1.1	71–89	70–124	0.22	0.20	0.047	0.010	0.21
171 Pup	F9V	15.27	1.57	79–99	78–137	0.21	0.24	0.046	0.012	0.26
bet TrA	F1V	12.42	8.71	228–285	226–397	0.23	0.39	0.046	0.016	0.35
alf Cen A	G2V	1.35	1.66	918–1147	908–1596	0.18	0.24	0.045	0.012	0.27
20 LMi	G3VaHde	14.92	1.37	75–94	75–131	0.21	0.22	0.045	0.011	0.24
b Her	F6V	15.76	2.02	87–108	86–151	0.21	0.27	0.045	0.013	0.28
tau01 Hya	F5V	18.24	4.05	106–132	105–184	0.21	0.34	0.043	0.016	0.36
rho01 Cnc	K0IV-V	12.58	0.635	61–76	60–106	0.19	0.10	0.043	0.005	0.11
HD 176051	F9V+K1V	14.89	1.42	77–96	76–134	0.20	0.20	0.042	0.010	0.24

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
HD 69830	G8:V	12.58	0.609	60–74	59–104	0.19	0.08	0.042	0.004	0.09
HD 160346	K3-V	10.71	0.32	51–63	50–88	0.18	0.04	0.041	0.002	0.05
gam CrA	F8V+F8V	17.89	4.8	118–147	116–204	0.19	0.30	0.041	0.014	0.34
HD 166620	K2V	11.09	0.363	52–65	52–91	0.17	0.04	0.037	0.002	0.05
alf CMi	F5IV-V+	3.50	7.21	737–921	729–1282	0.18	0.34	0.036	0.014	0.40
tau06 Eri	F5IV-V	17.78	5.15	123–153	121–213	0.17	0.25	0.035	0.011	0.32
gam CrA A	F8V	17.03	2.47	89–111	88–154	0.17	0.20	0.035	0.009	0.26
10 UMa	F3V+G5V	18.79	6.73	132–166	131–230	0.17	0.27	0.034	0.011	0.32
ksi Oph	F2V	17.50	4.35	114–143	113–199	0.16	0.24	0.033	0.011	0.33
rho Gem	F1V	17.83	5.44	126–157	124–218	0.15	0.23	0.032	0.010	0.32
HD 172051	G6V	13.02	0.682	61–76	60–106	0.14	0.04	0.031	0.002	0.07
70 Vir	G4Va	18.09	3.1	93–117	93–163	0.14	0.18	0.029	0.008	0.29
phi02 Cet	F7V	15.92	1.79	81–101	80–140	0.14	0.13	0.029	0.006	0.21
g Lup	F3/5V	17.39	3.33	101–126	100–175	0.14	0.18	0.029	0.008	0.29
9 Pup	G0V	16.67	1.95	80–101	80–140	0.14	0.12	0.028	0.006	0.20
HD 188088	K3VaCN1	14.14	0.803	61–76	60–106	0.13	0.03	0.028	0.001	0.04
58 Oph	F5V	17.17	2.72	92–115	91–160	0.13	0.14	0.027	0.007	0.25
HD 3443	K1V+G	15.53	1.32	71–89	70–123	0.13	0.08	0.027	0.004	0.13
mu. Vir	F2V	19.13	8.32	145–181	143–252	0.13	0.23	0.027	0.009	0.34
i Boo	G0Vn	12.95	0.639	59–74	59–103	0.12	0.02	0.026	0.001	0.03
nu.02 Lup	G3/5V	14.74	1.04	66–83	66–115	0.12	0.04	0.026	0.002	0.06
pi.01 UMa	G0.5V	14.43	0.972	66–82	65–114	0.12	0.03	0.025	0.001	0.05
eta Crv	F2V	18.24	4.77	115–144	114–200	0.12	0.16	0.025	0.007	0.30
psi Vel	F3VFe-0	18.56	9.87	163–203	161–283	0.11	0.20	0.023	0.008	0.33
51 Peg	G2IV	15.51	1.38	73–91	72–126	0.11	0.07	0.023	0.003	0.13
psi05 Aur	F9V	16.61	1.82	78–97	77–136	0.11	0.07	0.023	0.003	0.13
HD 115404	K2V	10.98	0.302	48–60	48–84	0.10	0.00	0.022	0.000	0.00
tet Cyg	F3+V	18.42	4.27	108–135	107–187	0.10	0.13	0.021	0.006	0.29
HD 37394	K1	12.26	0.477	54–68	53–94	0.10	0.01	0.021	0.000	0.01
i Cen	F2V	19.24	5.87	121–151	120–210	0.10	0.14	0.021	0.006	0.28
HD 5015	F9V	18.90	3.62	97–121	96–168	0.09	0.10	0.021	0.005	0.23
del Gem	F2VkF0m	18.60	10.6	168–210	166–292	0.10	0.17	0.019	0.006	0.33
alf Cha	F5VFe-0	19.48	7.11	131–164	130–229	0.09	0.14	0.019	0.006	0.31
HD 74576	K3V	11.19	0.318	48–60	48–84	0.09	0.00	0.019	0.000	0.00
HD 114837	F6VFe-0	18.23	2.99	91–114	90–158	0.08	0.08	0.017	0.003	0.21
HD 103932	K4+V	10.17	0.212	43–54	43–76	0.07	0.00	0.016	0.000	0.00
eta Cru	F2V	19.67	6.66	126–157	125–219	0.08	0.11	0.016	0.005	0.29
HD 207129	G2V	15.55	1.21	68–85	67–118	0.07	0.01	0.016	0.000	0.02
6 Cet	F8VFe-0	18.88	3.3	92–116	91–161	0.07	0.06	0.014	0.003	0.19
HD 217357	K7+Vk	8.23	0.101	37–46	37–64	0.07	0.00	0.014	0.000	0.00
e Vir	G0V	17.57	2.19	81–101	80–141	0.07	0.03	0.014	0.001	0.10
tau Cyg	F2IV	20.16	10.1	151–189	150–263	0.07	0.12	0.013	0.004	0.34
ksi Gem	F5IV-V	18.42	12.5	184–230	182–321	0.06	0.10	0.012	0.004	0.32
tet Dra	F9V	21.51	9.21	135–169	134–236	0.06	0.09	0.012	0.004	0.31
tau PsA	F6V	18.45	2.95	89–112	88–156	0.05	0.03	0.011	0.002	0.14
15 LMi	G0.5Va	18.80	2.77	85–106	84–148	0.05	0.03	0.011	0.001	0.11
HD 114613	G3V	20.45	4.29	97–122	96–169	0.05	0.05	0.010	0.002	0.20

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